

Akhmad Nizar Zakaria

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Professional Summary

Machine Learning Engineer and Data Scientist with 1 year of experience in developing end-to-end machine learning solutions, from ETL to model deployment in Azure environments. Proficient in Python, Azure Machine Learning, and data pipelines. Specialized in building models for marketing applications, including campaign budget allocation, performance optimization, and creative attribute analysis using computer vision and video indexing. Certified Azure Machine Learning Associate with a proven track record of enhancing business efficiency through data-driven solutions.

Technical Skills

- **Programming Languages:** Python, C++, C
 - **ML/AI Libraries:** Scikit-learn, Pandas, TensorFlow, OpenCV, XGBoost, CatBoost
 - **Tools & Platforms:** Azure ML Platform, Azure AI Foundry, Azure Video Indexer, Azure Custom Vision, Azure AI Vision, Google Cloud Vision API, Streamlit
 - **Big Data & Cloud:** Azure (Blob Storage), Google Cloud Storage
 - **Model Development:** Regression, Classification, Clustering, Computer Vision, NLP
 - **Databases:** Google Cloud BigQuery, Azure PostgreSQL
 - **DevOps & Tools:** Azure Functions, Azure ML Platform
 - **Methodologies:** Agile, CRISP-DM, Kanban, Scrum
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Work Experience

ML Engineer

Devdat, Jakarta, Indonesia [Remote]
May 2024 – Present

As a Machine Learning Engineer and AI Data Scientist, I specialize in developing end-to-end AI solutions, from advanced RAG systems to high-precision forecasting models. I focus on optimizing large-scale data pipelines and implementing production-grade Generative AI applications within the healthcare and marketing analytics domains.

Key Responsibilities & Achievements:

Generative AI & Clinical Intelligence

- Developed an automated system to extract data from Clinical Research Forms (eCRF) into structured JSON formats optimized for HTML rendering.
- Architected a Clinical Trial Matching System (CTMS) for oncology utilizing Retrieval-Augmented Generation (RAG) and SLM/LLM reasoning to optimize patient-trial alignment.
- Built a context-aware recommendation engine for Adverse Events (CTCAE) using RAG and LLMs to provide precise clinical reasoning.
- Conducted modality support assessments (Text, Image, Audio, Video) for various frontier models like Gemini, GPT, and Claude to ensure platform compatibility.

Predictive Modeling & Marketing Science

- Engineered business-grade time-series forecasting models with integrated output guardrails and probabilistic ranges (P10, P50, P90) to ensure realistic business outputs.
- Optimized forecasting accuracy for high-seasonality periods, specifically addressing Ramadhan impacts through hyperparameter tuning and seasonality adjustments.
- Developed machine learning models for marketing budget allocation using regression and Bayesian optimization to maximize cross-channel performance.

MLOps & Analytics Infrastructure

- Built an internal cost calculation engine to estimate token, video, and storage costs for Azure AI services, providing granular usage transparency.
 - Implemented video content analysis and creative deduplication using Azure Video Indexer and Computer Vision APIs to optimize engagement insights.
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Education

Bachelor of Computer Engineering

Brawijaya University, Malang, Indonesia

September 2020 - January 2024

- GPA 3.73/4.0
 - Thesis: Vowel Letter Reading Training System for Children using Fast Fourier Transform Features and Random Forest Algorithm Based on Arduino Uno.
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Personal Projects/Portfolio

Cafe Recommendation System

[Github](#) | 2023

- Developed a content-based filtering recommendation system for cafes in Malang, leveraging user preferences such as ambiance, price, and amenities (e.g., charging ports, spacious parking, industrial theme).
- Enabled search functionality allowing users to find cafes using descriptive queries, such as “cafes with charging ports and spacious parking lots.”
- Collaborated with cloud and mobile development teams to integrate the system into a mobile application.

Crypto and Stock Market Screener with AI Consultancy and Sentiment Analysis

[Github](#) | 2023

- Built a Streamlit-based dashboard to display real-time stock and cryptocurrency market charts alongside sentiment analysis for selected assets.
 - Integrated an AI agent to analyze market charts and provide actionable insights and recommendations based on sentiment and trends.
 - Designed the system to process and visualize real-time data, enhancing decision-making for users.
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Certifications

- Azure Machine Learning Associate, Microsoft, 2026

- TensorFlow Developer Certificate, Google, 2026
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Languages

- Indonesian (Native)
- English (Upper Intermediate)